Algorithm to Analyze Tennis Serve Form

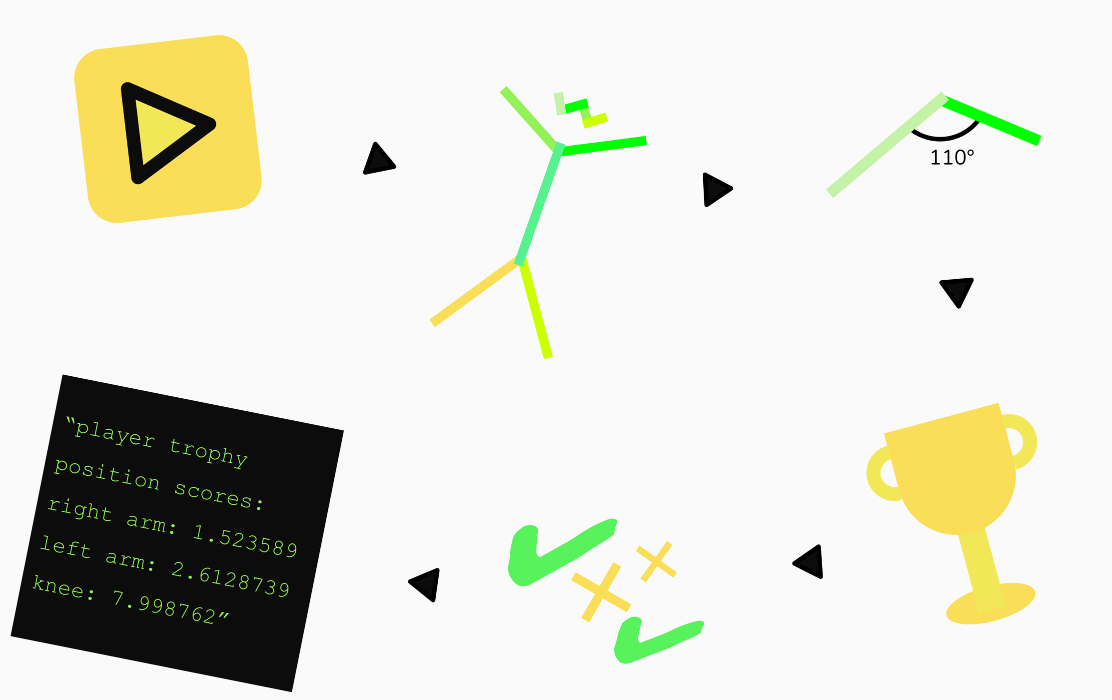
Image of tennis ball created by researcher using Google Drawings.

Introduction

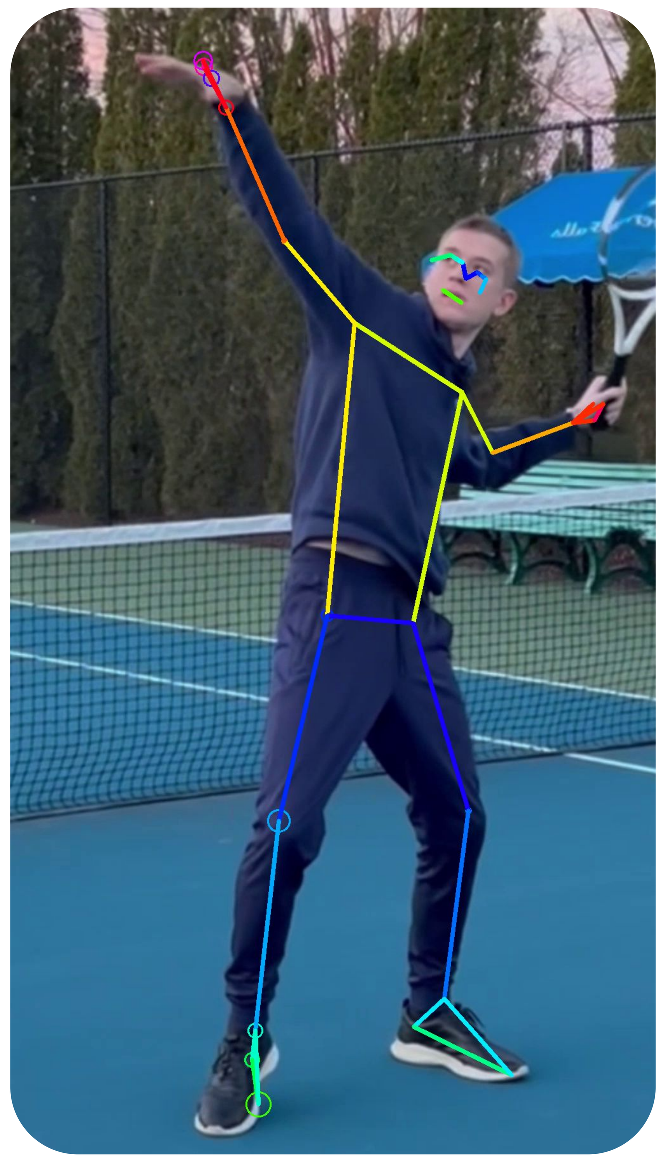
This project gives a tennis player feedback on their slice serve. The player **inputs a video** of them doing a slice serve. Using movement data from the video, an algorithm **grades** the form of the slice serve. The program **outputs** one or more **scores** or **verbal feedback** on their form.

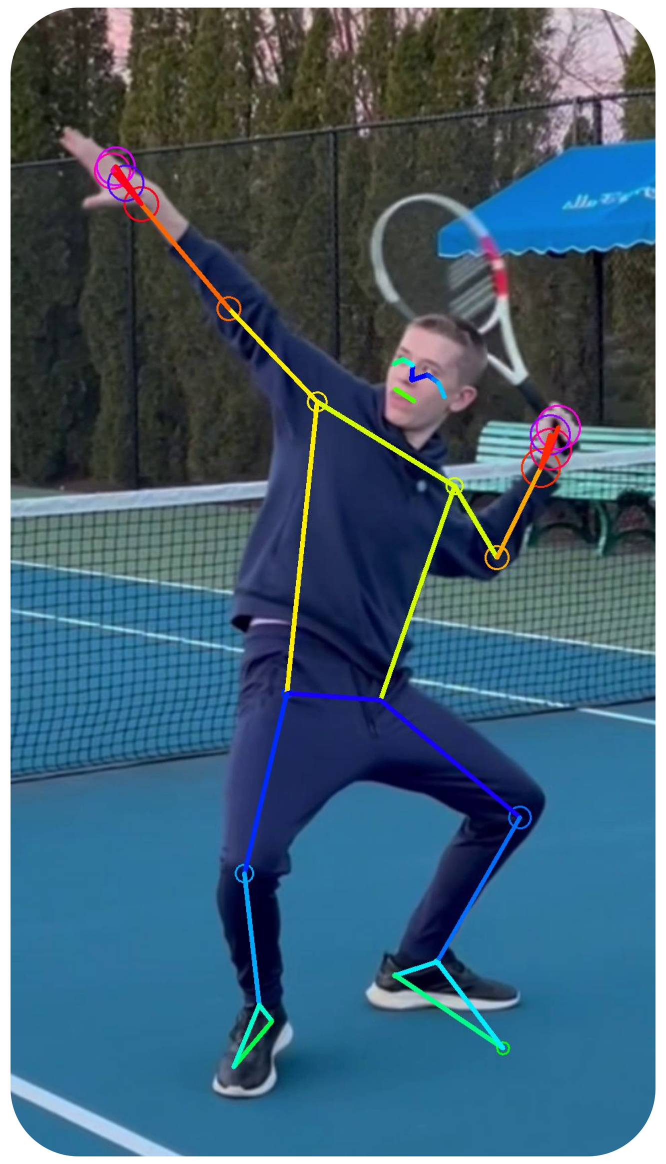
Mechanism

- Motion capture w/ computer vision model
- Keypoints → **"joint angles"**
- Function detects frame where player is in **trophy position**
  - Trophy position (wind-up) quality is a good indicator of serve quality
- Individual scores: **left elbow angle, left arm angle of inclination, right elbow angle, right elbow-shoulder alignment, and knee angles**
  - Each score between 0 and 1
  - Weighted and unweighted scores
- Large language model
  - Input: individual scores + prompt
  - Output: **verbal feedback**



Explanation diagram created by researcher in Google Drawings.





Images depict detected **trophy position frames** from videos of self doing a slice serve. Scores from left to right: 0%, 20% (straight left elbow angle), 57% (improved trophy position)

Three photographs taken by researcher on iPhone 13 Pro.

	THETIS Beginners	THETIS Advanced	THETIS Overall	Self-Testing Video	Roger Federer Video
Average Unweighted Score	0.951	1.864	1.350	4.265	5.474
Average Weighted Score	23.6%	62.7%	40.7%	19.5%	46.9%

Average Unweighted vs. Weighted Overall Scores for Sets of Videos

Note: unweighted scores are likely a better indicator of quality than weighted.

Data table created by researcher in Google Drawings.

References

Bazarevsky, V. et al. (2020). *Blazepose: On-device real-time body pose tracking* (arXiv:2006.10204). arXiv. https://doi.org/10.48550/arXiv.2006.10204

Hill, G. (2014, August 20). The serve part 2: Benoit paire and the power of the trophy position. Tactical Tennis. https://www.tacticaltennis.com/serve-part-2-benoit-paire-power-trophy-position/

Kovacs, M., & Ellenbecker, T. (2011). An 8-stage model for evaluating the tennis serve: implications for performance enhancement and injury prevention. *Sports health*, 3(6), 504–513. https://doi.org/10.1177/1941738111414175

Qwen Team. (2025). *Qwen3 technical report* (arXiv:2505.09388). arXiv. https://arxiv.org/abs/2505.09388

Methods

The project was created on a computer running Windows 11. It used **Python 3.9** and **CUDA 12.9**.

The CV model **Blazepose Fullbody** was used to derive movement data from video. The file running the model was modified in order to derive some extra data, create functions for making visualizations, and organize mass runs on sets of videos.

The algorithm was developed manually based on **slice serve research**. The scoring reflects the handful of simple factors in optimizing the trophy position.

The LLM, **Qwen3:4b**, was chosen because it is powerful but free-to-use and locally runnable. The model is prompted differently based on available data and scores.

Conclusions

AI Use in this Project

Artificial intelligence was used to derive movement data and convert scores into verbal feedback.

AI is not necessary to evaluate form; it is generally more efficient to use a conventional algorithm.

Automatic Form Evaluation

Slice serve form can be evaluated automatically. This may well hold true for other strokes.

Product Viability

This project can realistically compete with some features of paid products.

Next Steps

- Further **refine** slice serve **grading**
- Work on **other strokes**
- Create an application **UI**